Nuclear Fission Mini Project

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1 One-Dimensional Model

Q5.1 A one-dimensional model was made in python by using a for loop with 100 initial fissions which have a random starting position defined as the length * a random number from 0 to 1 this gives a random position along the length for the initial fission. A second loop then simulates the secondary neutrons released and gives them a random direction which determines wether they stay within the length to then produce a secondary fission.

Q5.2 An updated version of this code using the neutron function was made by simply using the function within the range of the secondary for loop that determines the number of secondary neutrons produced. In this step I also made the 1D simulation a function so it can be used later.

Q5.3 by using the function within a loop which finds a number of secondary fissions for each simulation a list of Secondary/initial Ratios can be found to find the critical point in the simulation.

100 Initial Fissions		**10000 Initial Fissions**	
L **(cm)**	**Ratio**	L **(cm)**	**Ratio**
10	0.400	10	0.376
11	0.610	11	0.581
12	0.720	12	0.748
13	0.820	13	0.877
14	1.020	14	0.991
–	–	15	1.093
14 cm is Critical Length		**15 cm is Critical Length**	

Table 1: Comparison of Secondary/Initial Ratios

This table shows the values obtained up to the critical ratio of 1 when using 100 initial Fissions the critical length found was in the range of 14-16 cm in this case it was 14, however when usign 10000 initial fissions the values was consistent at 15cm this suggests that it is a more viable estimate for the critical length.

2 Three-Dimensional Model

Q5.4 updating the model to 3D involved a starting position using x y and z coordinates with a random direction using phi and theta as angles instead of just left and right.

Q5.5 By using the 3D function with 1000 initial fissions (any more than 1000 took a while to compute) gave a value for the critical length of 15 cm which can be used along with the density of uranium to find a critical mass as shown below.

Length L (cm)	**Secondary/Initial Ratio**
10	0.574
11	0.704
12	0.781
13	0.846
14	0.942
15	1.054
15 cm is Critical Length	

Table 2: Results for 1000 Initial Fissions

Critical Mass Estimate:
Critical mass = 63.112 kg

In order to find a more accurate value than above 100 values between 14-15cm were tested, this allows us to find the critical length to 2 decimal places for a more accurate final critical mass.

Increased Accuracy Results:
$L = 14.29$ cm is Critical Length
Increased accuracy critical mass = 54.568 kg